

Detecting similar book reviews

Elisa Rosa and Mary Jane Reyes

September 12, 2025

Abstract

This project addresses the problem of detecting similar book reviews in the Amazon Books Review dataset. The analysis combines text preprocessing, shingling, MinHash signatures with Locality-Sensitive Hashing (LSH) to efficiently identify candidate pairs of reviews. Results show that the approach can successfully detect both near-duplicate and moderately similar reviews, while maintaining scalability when applied to larger data samples. These findings confirm the effectiveness of MinHash and LSH for large-scale similarity detection in textual datasets.

1 Introduction

In this project we address the problem of detecting similar book reviews using the "Amazon Books Review" dataset, that is downloaded dynamically via the Kaggle API.

2 Data overview

The dataset used for this project contains around 3 million reviews of books available on Amazon. To confirm the dataset size we detected the number of rows in the file, which turned out to be exactly 3.000.000 rows.

The dataset contains several columns describing both the books and the associated reviews. Table 1 provides an overview of the variables together with a brief description of their content.

Variable	Description
Id	Unique identifier of the book
Title	Title of the book
Price	Price of the book
User_id	Anonymized unique identifier of the reviewer
profileName	Display name of the reviewer
review/helpfulness	Helpfulness votes in the form "x/y", where x is the number of users who found the review helpful and y the total number of votes
review/score	Rating score given by the reviewer (from 1.0 to 5.0)
review/time	Unix timestamp of when the review was posted
review/summary	Short textual summary of the review
review/text	Full text of the review

Table 1: Description of the variables contained in the Amazon Books Review dataset

3 Pre-processing

3.1 Dataset sampling

Given the large size of the dataset, we decided to initially work on a smaller subset to speed up development and testing. We implemented a global flag, which allows us to choose between loading the full dataset or only a sample. For our work, we loaded only the first 20.000 rows.

3.2 Column selection and missing values

After that, we selected only the columns relevant to our analysis: the book identifier (**Id**), the book title (**Title**) and the full review text (**review/text**). These are the only fields needed to compute textual similarity between reviews.

We verified that the dataset does not contain any missing values in these columns. In particular, we checked that every row had both a valid book ID and an associated review text. This allowed us to proceed without needing to clean or impute missing data at this stage.

3.3 Removal of HTML entities

We also included a step to clean the text from HTML entities (such as `"` or `&`). This was done by decoding these entities into their corresponding characters, ensuring that the reviews contained plain and readable text. This step prevents that two very similar reviews are considered different simply because one contains encoded entities while the other does not.

3.4 Text Processing

In order to compare different reviews based on their textual content, we first applied a series of pre-processing steps to clean and normalize the raw text. These transformations help reduce noise and improve the quality of similarity computations.

We defined a pre-processing function that takes a review as input and returns a list of clean, meaningful tokens. The function includes the following steps:

- **Lowercasing:** all characters in the text are converted to lowercase to avoid treating words like *Book* and *book* as different tokens.
- **Punctuation removal:** all punctuation symbols are removed using a regular expression, except for apostrophes, which are kept to preserve contractions (e.g., *don't*, *it's*).
- **Tokenization:** the cleaned text is split into individual words using whitespace as a delimiter.
- **Stopword removal:** common English words that carry little semantic meaning (such as *the*, *and*, *is*) are removed from the token list. We initially started from a file that contained a list of stopwords provided among the course materials; however, we noticed that some words in the original list, such as *not*, could significantly alter the meaning of a review if removed. For example, removing negations might incorrectly increase the similarity between two reviews that express opposite sentiments. For this reason, we decided to manually reduce the list to avoid over-filtering and retain important contrastive words.

After applying these steps, each review is represented as a list of normalized tokens. These tokens will be used to generate shingles in the next phase of the pipeline.

4 Shingles generation

After pre-processing the text into clean tokens, we converted each review into *shingles*. For our analysis, we chose $k = 2$, meaning that each review was represented as a set of all its 2-word combinations. This representation captures local word context and provides a better notion of similarity than using individual words alone.

As an example, the sentence:

“This is only for Julie Strain fans.”

would produce the following shingles:

`{this is, is only, only for, for julie, julie strain, strain fans}`

Using this method, each review was transformed into a set of unique consecutive word pairs. This representation captures local word context and serves as the basis for computing similarity between reviews using set-based measures like the Jaccard index.

5 Similarity Detection

5.1 MinHash signatures

Once each review was represented as a set of word-level shingles, we applied MinHashing to approximate the Jaccard similarity between reviews. The idea behind MinHash is to create a compact signature for each set such that the similarity between two signatures approximates the true Jaccard similarity of the original sets.

We used $n = 100$ independent hash functions of the form $h(x) = (a \cdot x + b) \bmod p$, with randomly chosen parameters a and b and a large prime $p = 4294967311$. We fixed the random seed before generating the hash parameters.

For each review, we computed a MinHash signature consisting of 100 values — one for each hash function — by taking the minimum hashed value over all shingles in the set. Reviews with no shingles were excluded from the next steps.

5.2 Locality-Sensitive Hashing (LSH)

To avoid comparing every pair of reviews, which would be computationally expensive, we implemented Locality-Sensitive Hashing (LSH) to identify candidate pairs likely to be similar.

Each 100-element MinHash signature was divided into 25 bands of 4 rows each. For each band, the corresponding subvector was hashed into a bucket. Two reviews that fall into the same bucket in at least one band were considered candidate pairs.

This approach dramatically reduces the number of comparisons needed, as only reviews sharing a bucket are considered for similarity. At the end of this process, we obtained a set of 1.327 candidate pairs.

5.3 Similarity filtering and final pairs through Jaccard similarity

Once all candidate pairs were identified using LSH, we computed the approximate Jaccard similarity for each of them using their MinHash signatures. This approximation is based on the proportion of matching elements between two signatures.

Given two MinHash signatures of length 100, the similarity score is simply the number of matching positions divided by 100. This provides an efficient way to estimate how similar two sets of shingles are, without computing the full Jaccard coefficient directly.

We then applied a threshold range to filter review pairs, selecting those with an approximate similarity between 0.5 and 0.95. In this way we were able to capture both reviews that are fairly similar and those that are almost identical.

As a result, we identified 111 pairs of reviews within this interval, ranging from moderately similar to almost identical.

To further verify the quality of the retrieved matches, we selected a sample of 10 pairs from the range between 0.5 and 0.95 similarity, as shown in Table 2. The inspection showed that the pipeline was able to capture both reviews with substantial but not identical overlap, as well as pairs that were nearly identical. This confirms that the MinHash + LSH approach can effectively identify not only duplicates, but also reviews that convey very similar content with different wording.

Review #i	Review #j	Similarity
9008	9026	0.84
8000	10088	0.95
7601	7760	0.95
9816	9817	0.84
3933	3945	0.61
9009	9161	0.93
15318	15788	0.90
2457	11606	0.73
7152	7188	0.91
15005	15835	0.95

Table 2: 10 random matched review pairs with similarity in the range [0.5, 0.95]

6 Results and Discussion

To provide a clearer picture of the qualitative results, we report below two examples of review pairs detected and showed in the previous chapter of the report.

The first one corresponds to a moderately similar case (similarity ≈ 0.61), where the reviews share much of the same content but differ in specific details and phrasing.

The second one represents a highly similar pair (similarity ≈ 0.95), where the overlap is nearly exact with only minor textual variations.

These examples illustrate how the MinHash + LSH approach can successfully identify both reviews that are not literal duplicates but still very close in meaning, and reviews that are almost identical.

Differences between the reviews are emphasized in both tables with italic and underline.

Pair 5	Similarity: 0.610
Title i	The Bruce Lee Story
Review i	I have read this book twice, what a great book! Anyone intersted in the masters life should definitely get this book. It is written by Mrs. Lee, the only person in the world who knew him best. There are great photos, and great detail about their lives together. I have been a fan for over 10 years, and this one is one of the best biographys I’ve ever read about the greatest martial artist, and philosipher who ever lived.
Title j	The Bruce Lee Story
Review j	I have read this book twice, what a great book! Anyone intersted in the masters life should definitely get this book. It is written by <i>his ex-wife</i> , the only person in the world who knew him best. There are great photos, and great detail about their lives together. I have been a fan for over <u>15</u> years, and this one is one of the best <i>biographies</i> I’ve ever read about the greatest martial artist, and philosipher who ever lived. <i>Highly recommended!</i>

Table 3: Example of two reviews with moderate similarity (Pair 5).

Pair 3	Similarity: 0.950
Title i	Foundation
Review i	You’ve already read the publisher’s summary for yourself, so let me just add that this is THE classic SF trilogy. Book one, "Foundation," creates the future world, its declining culture, and how a group of strong-minded individuals don’t want to see knowledge die. How they do that drives this novel. In book two, Asimov introduces the mystery of the mule, and in book three Asimov pulls a rabbit out the hat called the Second Foundation. If you enjoy books that entertain, books that offer good mysteries, and books that make you think about bigger things in life, than this series is for you. It’s too bad our political leaders don’t read novels like this. Maybe we’d avoid the continual messes they get countries in.
Title j	Foundation
Review j	<i>You can read</i> the publisher’s summary for yourself, <i>but</i> let me just add that this is THE classic SF trilogy. Book one, "Foundation," creates the future world, its declining culture, and how a group of strong-minded individuals don’t want to see knowledge die. How they do that drives this novel. In book two, Asimov introduces the mystery of the mule, and in book three Asimov pulls a rabbit out the hat called the Second Foundation. If you enjoy books that entertain, books that offer good mysteries, and books that make you think about bigger things in life, than this series is for you. It’s too bad our political leaders don’t read novels like this. Maybe we’d avoid the continual messes they get countries in.

Table 4: Example of two reviews with high similarity (Pair 3).

7 Scalability

To evaluate the scalability of our pipeline, we re-ran the entire analysis using a significantly larger sample of **100.000** book reviews, instead of the original 20.000. The objective was to verify whether the methodology, particularly the steps involving MinHash signature computation and Locality-Sensitive Hashing (LSH), remained reliable as the dataset size increased, even though not as computationally fast as the 20.000 sample.

The results confirmed that the system scales well. All components of the pipeline, from pre-processing and shingling, to signature generation and candidate pair extraction, continued to function correctly. In this extended run, the system produced **11.761 candidate pairs**, out of which **706 pairs** had an estimated similarity between 0.5 and 0.95. This demonstrates that the pipeline retains both efficiency

and accuracy even as the dataset grows substantially.

To further verify the quality of the retrieved matches, we selected a sample of 10 pairs from the range between 0.5 and 0.95 similarity, as shown in Table 5.

Review #i	Review #j	Similarity
88112	88159	0.93
22733	22748	0.66
36376	36489	0.95
17389	18330	0.94
87072	87520	0.88
66721	67560	0.90
18134	61844	0.94
62358	62601	0.90
8032	56581	0.93
96698	96734	0.73

Table 5: 10 random matched review pairs with similarity in the range [0.5, 0.95]

To illustrate the outcome of our pipeline on the larger sample, we report below an example of two reviews that were flagged as moderately similar, with an approximate Jaccard similarity of 0.73. Differences are highlighted in *italic* and underline.

Pair 10	Similarity: 0.730
Title i	An Isolated Incident
Review i	The book is great for simple reading , and I found it to be quite intresting through out . It is not "deep" and is good for light reading . The characters are quite intresting although not much emotion is <u>show</u> . It does give some insight to human behaviour and society's wrongful conduct . However the ending made me wonder if the author wanted to keep the option of a sequel . It could have ended more conclusively .
Title j	An Isolated Incident
Review j	The book is great for simple reading , and I found it to be quite intresting through out . It is not "deep" and is good for light reading . The characters are quite intresting although not much emotion is <u>shown</u> . It does give some insight to human behaviour and society's wrongful conduct . However the ending made me wonder if the author wanted to keep the option of a sequel . It could have ended more conclusively . <i>I would recommend anyone intrested in murder mystries to read it .</i>

Table 6: Example of two reviews with moderate similarity (Pair 10).

8 Conclusions

In this project we developed and tested a pipeline for detecting similar reviews in the Amazon Books Review dataset. Through pre-processing, shingling, MinHash and Locality-Sensitive Hashing, we were able to efficiently identify pairs of reviews with substantial textual overlap. The analysis confirmed that the method works well in detecting both exact duplicates and reviews that are highly similar but not identical.

By selecting a threshold we showed how the pipeline can capture reviews ranging from similar to nearly identical. Furthermore, scaling the analysis from 20.000 to 100.000 reviews confirmed the robustness and scalability of the approach, which maintained consistent results without loss of accuracy.

Overall, the project demonstrates the effectiveness of MinHash and LSH in handling large-scale textual data.

9 Declaration of Originality

We declare that this material, which we now submit for assessment, is entirely our own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of our work. We understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by us or any other person for assessment on this or any other course of study. No generative AI tool has been used to write the code or the report content.